



PROJECT OVERVIEW · V5.6.0 · 2026

RealizeOS

The self-hosted AI operations system for your business — a coordinated team of AI agents that knows your venture, remembers, and executes.

OPEN CORE · BSL 1.1

LOCAL-FIRST

MULTI-LLM

MCP-NATIVE

Businesses adopted AI. They didn't get an **AI operation**.

Small and mid-size ventures use chatbots as clever text boxes. The work around them — context, follow-through, approvals, memory — is still manual.



No memory

Every chat starts from zero. Voice, clients, decisions and past work must be re-explained.



One-shot answers

Chatbots reply; they don't run multi-step workflows, hand off between specialists or chase outcomes.



Wrong model, wrong cost

A flagship model formats a date; a cheap model drafts strategy. No one routes by task.



Lock-in & risk

Knowledge lives inside a vendor. No audit trail, no approval gates, no ownership.

The gap: owners need an operations layer — agents that share one knowledge base, act under governance, and belong to the business, not the model vendor.

An "AI employee" is six hard problems stacked on top of each other

6 Governance — who may do what, with approval & audit

5 Security — prompt injection, secrets, RBAC

4 Orchestration — missions, handoffs, retries

3 Model economics — 4 providers, cost/quality

2 Memory — what to recall, when, how cheaply

1 Business knowledge — structured, owned, portable

Each layer depends on the one below it. Get knowledge wrong and memory is noise; get memory wrong and orchestration drifts; skip governance and nobody can trust the output.

40+

CORE SUBSYSTEMS

8

MISSION STATES

7

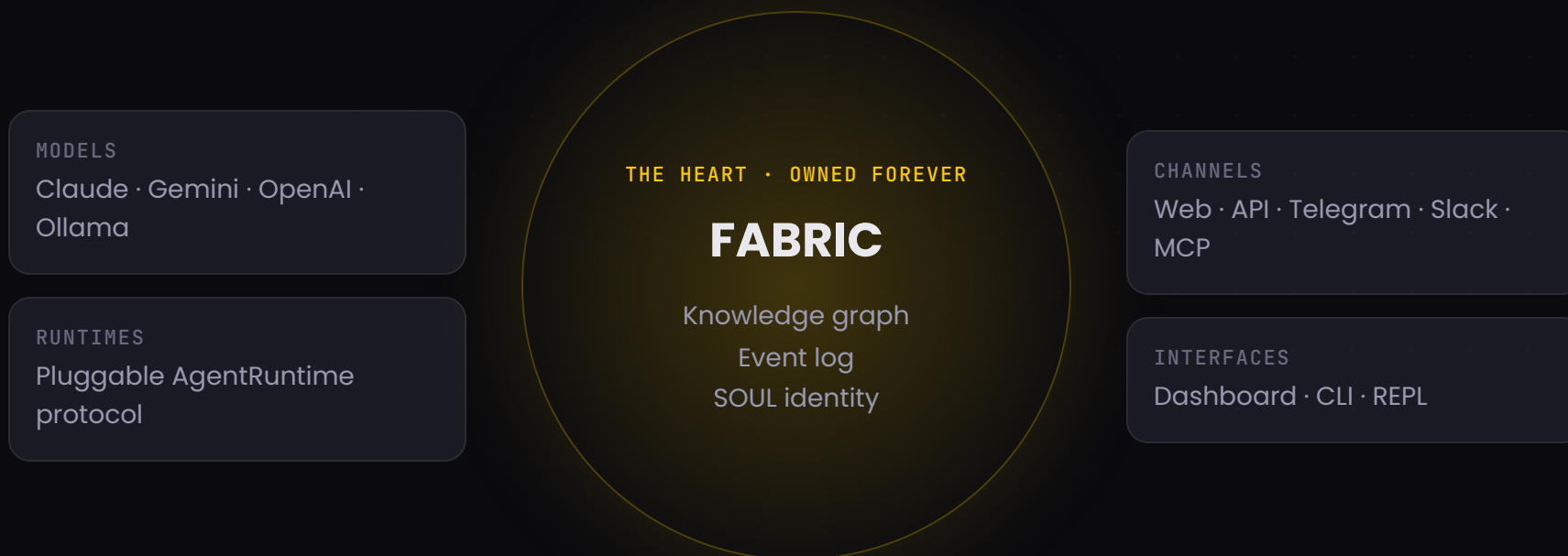
AGENT HANDOFF TYPES

5

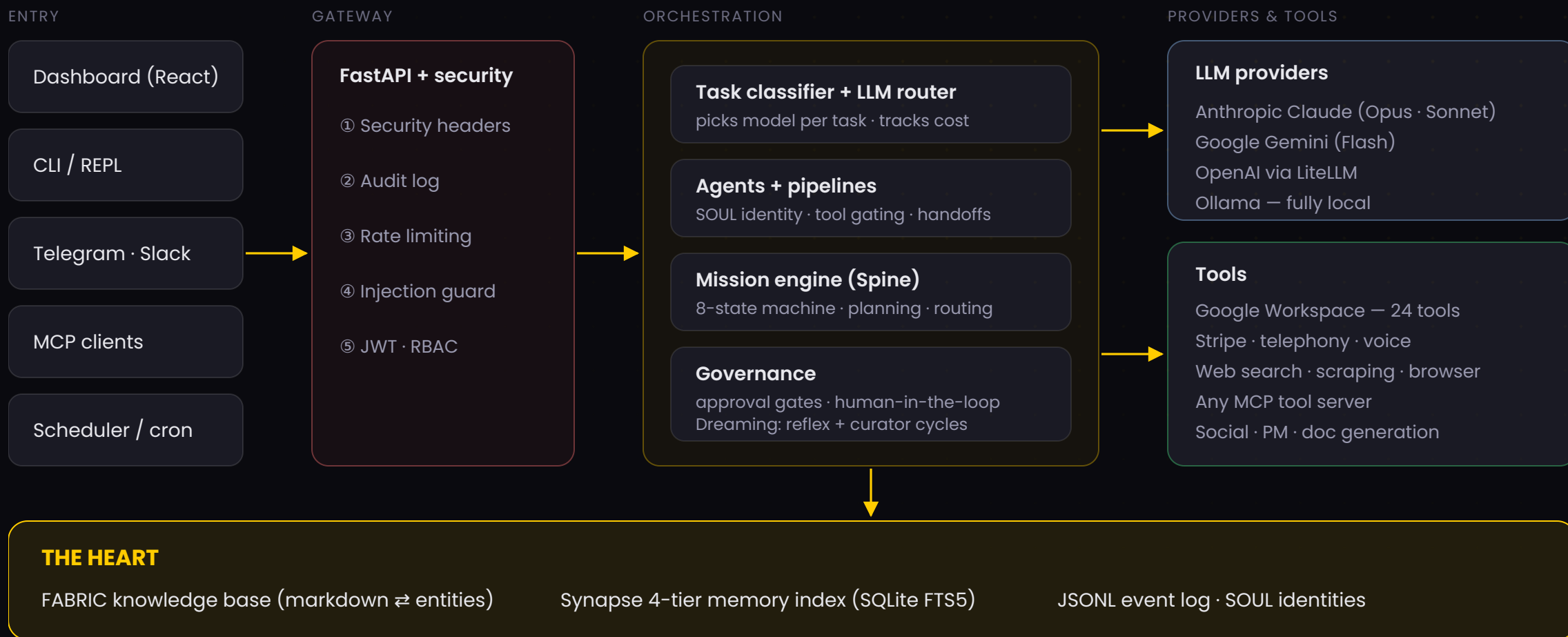
SECURITY LAYERS

You own the **Heart**. Everything else is a swappable adapter.

RealizeOS separates the durable asset — your knowledge, history and identity — from the fast-moving parts: models, agent runtimes, channels and even the UI.



How a request flows through the system



FABRIC structures the business. Synapse decides what the agent recalls.

FABRIC — SIX KNOWLEDGE LAYERS PER VENTURE

F

Foundations

Identity, voice, standards

A

Agents

AI team & routing guide

B

Brain

Domain & market knowledge

R

Routines

Skills, workflows, SOPs

I

Insights

Learnings, feedback, decisions

C

Creations

Drafts & deliverables

Plain markdown a human can edit — round-tripped into typed entities (contacts, decisions, commitments...) with provenance, trust scores and JSON-Schema validation.

SYNAPSE — 4-TIER MEMORY, CHEAPEST FIRST

L1 Hot TOC

Always in context — the map of what exists

L2 Full-text search

SQLite FTS5 over the knowledge base

L3 Tool catalog

Which capability to reach for

L4 Mission memory

State of long-running goals

Right model, right agent, right moment

Multi-LLM routing

Every task is classified, then sent to the cheapest model that can do it well.

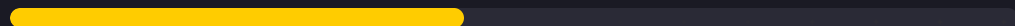
Simple · lookups, formatting

Gemini Flash



Content · writing, analysis

Claude Sonnet



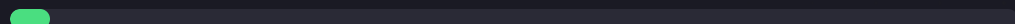
Complex · strategy, reasoning

Claude Opus



Private · sensitive data

Ollama (local)



BAR = RELATIVE CAPABILITY / COST TIER (ILLUSTRATIVE)

Composable agents

Scope, inputs, outputs, guardrails and allow-listed tools. Hot-reloaded from disk. Each carries a persistent SOUL: role, personality, expertise.

Pipelines & 7 handoff types

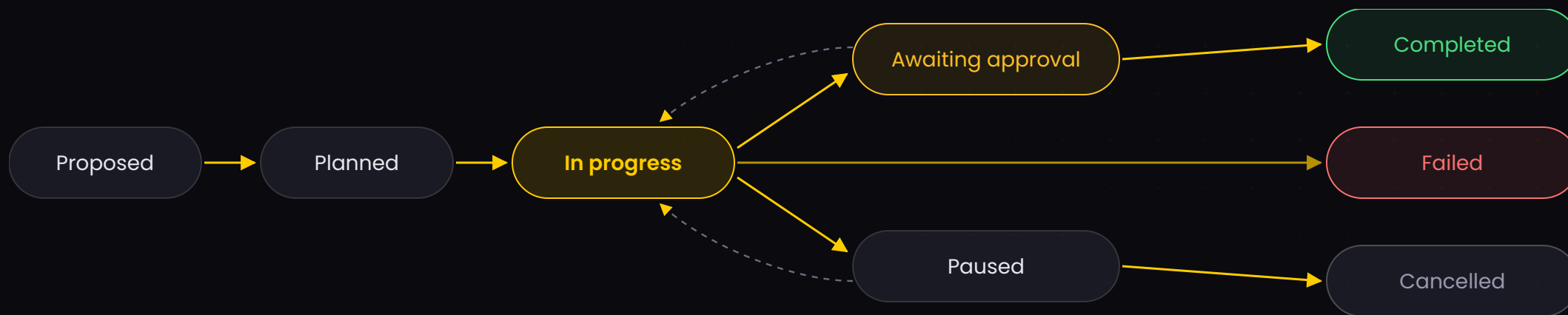
Standard, QA-pass / QA-fail, escalation, phase-gate, sprint, incident — with Dev → QA retry loops.

Dreaming

Background reflex & curator cycles propose knowledge updates into a Dream Inbox — a human approves before anything becomes "truth".

Goals, not prompts: an 8-state machine that sees work through

A mission is planned into steps, each routed to the best runtime and model, cost-tracked, and paused for approval where it matters.



per-step: runtime routing · model selection · € cost ledger · event-log entry (dashed = resume)

€0.028

TRACKED COST · 3 DEMO MISSIONS

JSONL

APPEND-ONLY EVENT LOG PER MISSION

HITL

APPROVAL GATES ON CONSEQUENTIAL ACTIONS

Boring where it should be, modern where it matters

BACKEND

- Python 3.11+
- FastAPI · Uvicorn
- httpx (async)
- Typer CLI

AI LAYER

- Anthropic SDK
- Google GenAI
- LiteLLM (OpenAI & more)
- Ollama (local)

DATA

- Markdown FABRIC files
- SQLite + FTS5 index
- JSONL event log
- JSON-Schema entities

FRONTEND

- React 19 · TypeScript 5.9
- Vite 8 · Tailwind 4
- TanStack Query
- @realizeos/design-system

PROTOCOLS

- MCP server (24 tools)
- HTTP + SSE streaming
- REST FABRIC API

SECURITY

- JWT · API keys
- 6-role RBAC
- Injection scanner

DISTRIBUTION

- Docker (GHCR)
- PyPI · npm CLI
- One-line installers

QUALITY

- pytest suite · CI
- Commitlint · Renovate
- ADRs & audit playbook

Trust is a feature: every request passes five gates

① **Security headers**

② **Audit logging** — JSONL + live SSE

③ **Rate limiting**

④ **Injection guard** — pattern + heuristic +
Unicode

⑤ **JWT auth + RBAC**

SIX ROLES, LEAST PRIVILEGE

OWNER

ADMIN

OPERATOR

USER

VIEWER

GUEST

- Human-in-the-loop approvals for payments, emails and other consequential actions
- Per-agent tool allow- and deny-lists
- Secrets redacted from errors and logs
- Self-hosted: data never has to leave your server

A production-scale codebase, not a prototype

54.5k

LINES OF PYTHON (CORE + API)

11.5k

LINES OF TSX DASHBOARD

2,032

AUTOMATED TESTS · 99 FILES

40+

CORE MODULES

24

MCP TOOLS · 4 FAMILIES

4

LLM PROVIDERS

12

BUSINESS TEMPLATES

5.6

CURRENT RELEASE

CODE DISTRIBUTION (APPROX. LINES)



■ Python backend 82.5% ■ TypeScript dashboard 17.5%

Free core. Paid expertise. No per-seat tax.

Open-source under BSL 1.1 lowers adoption friction to zero; revenue comes from making it work for a specific business.

TIER 0 • FREE

Open core

Self-host via Docker, pip or npx. Builds community, trust and a funnel.

TIER 1 • SETUP

Guided installation

Onboarding sessions: venture modelling, FABRIC population, agents & integrations wired.

TIER 2 • RECURRING

Maintenance + vertical consulting

Setup-plus-maintenance pricing: recurring revenue without SaaS lock-in.

TARGET VERTICALS — SHIPPED AS READY TEMPLATES

CONSULTING

AGENCY

SAAS

ECOMMERCE

ACCOUNTING

COACHING

FREELANCE

PORTFOLIO

REAL-ESTATE

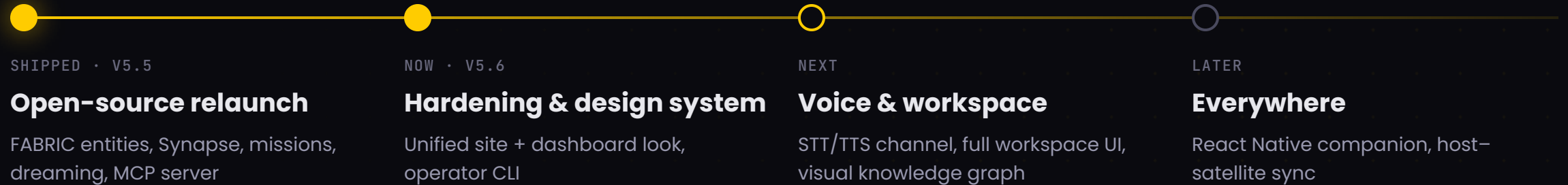
REAL-ESTATE DEVELOPER

PROPERTY MGMT

ARCHITECTURE FIRM

Why it wins: owners keep their data & knowledge; model costs drop through routing; each vertical template is reusable IP that shortens every new install.

Where it goes next



Your business knowledge is the asset.

RealizeOS makes it work.

github.com/SufZen/RealizeOS-5 · realization.co.il